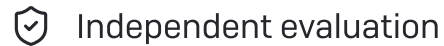
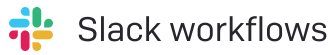
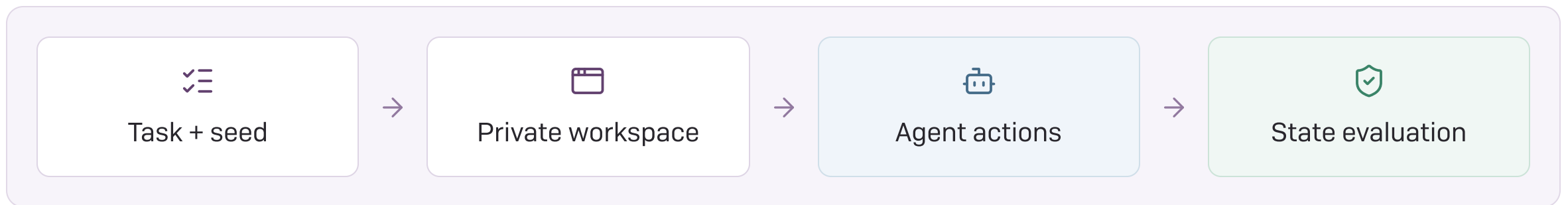


A Slack-like environment for computer-use agents.



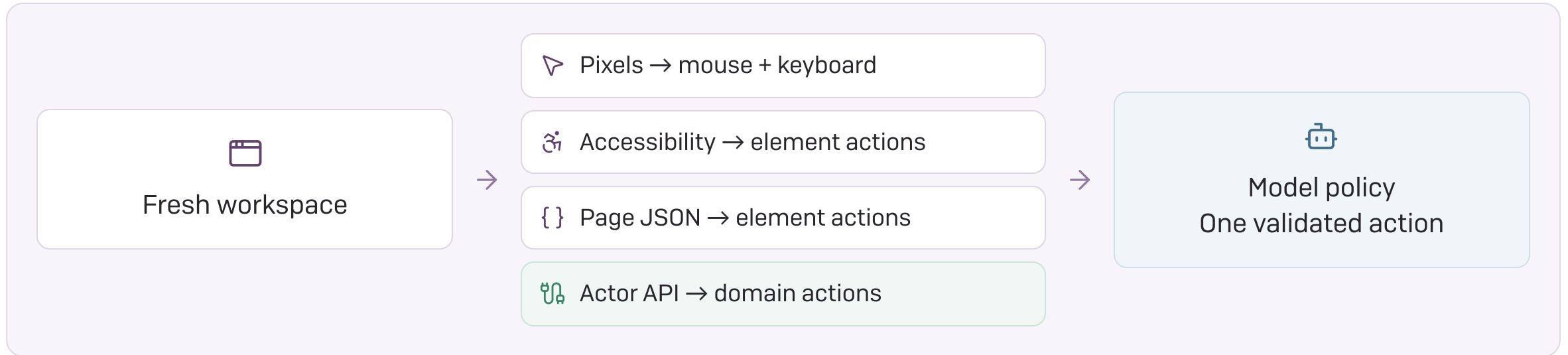
- Runnable web application with persistent, synthetic workspace state.
- 18 tasks covering search, threads, edits, DMs, pins, saved messages and channel details.
- Independent terminal-state evaluation and inspectable agent trajectories.



Implementation scope and prior work

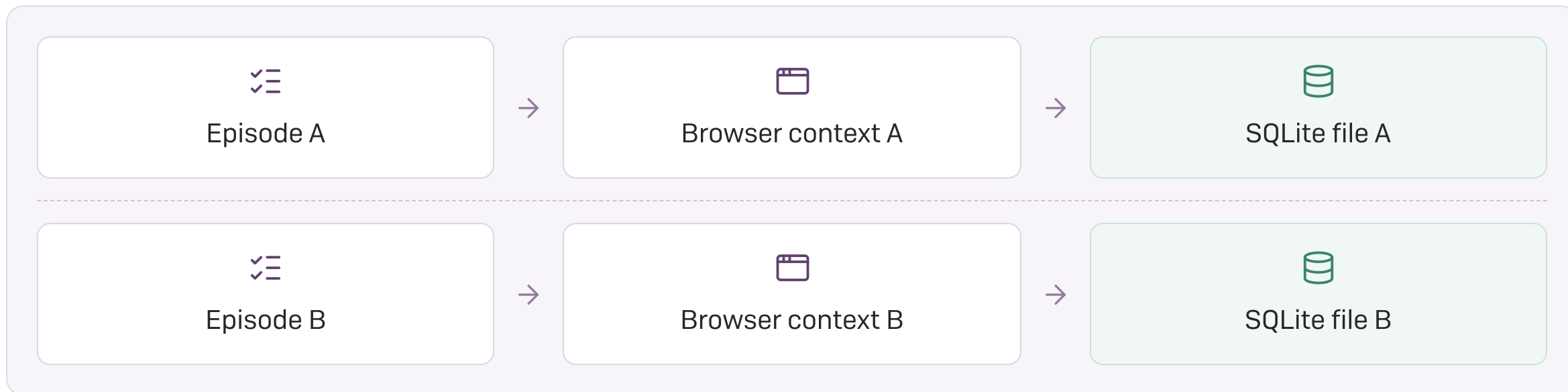
- React UI, deterministic transition engine, SQLite persistence and Playwright runner.
 - Static Slack mockups supply appearance, but not reset, reward or durable outcomes.
 - Cua Slack, env0 and Agent-Diff already address related environments; this is not a novelty claim.
 - Original focused implementation keeps task contracts and failure traces easy to inspect.
-
- MIT for original code; third-party asset restrictions remain separate.
 - Source review, not comparative execution: [prior-work survey and pinned references](#).

Four observation and action interfaces



- The actor never receives grader answers or control credentials.
- API access changes information and action granularity; it is not a computer-use score.
- Optional documentation and bounded history are explicit conditions, not hidden prompt changes.

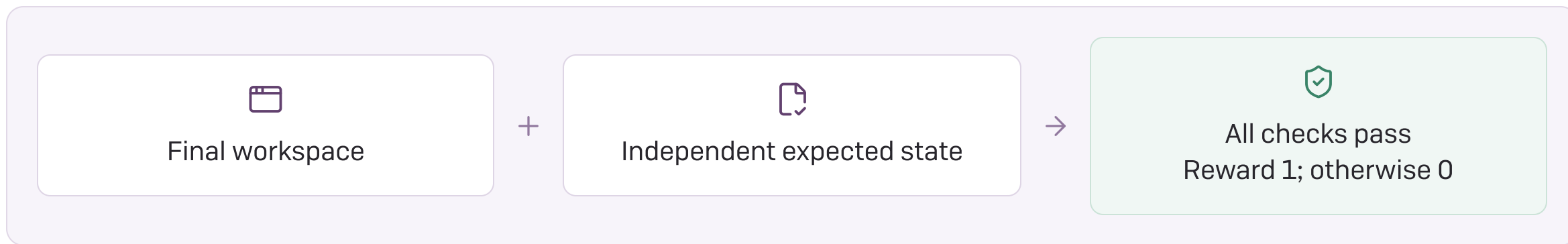
Episode lifecycle and session isolation



- Fresh contexts and private files prevent cross-episode state contamination.
- Transactional writes, revisions and request IDs enforce atomicity and idempotency.
- Actor and control listeners share a process: this is data isolation, not a hostile-code sandbox.

Tasks require correct state, not a “done” response

- Six control tasks; 12 deeper workflows with 3–6 coordinated mutations.
- Decision record: retrieve final approval, update description, pin the root, save the reply and acknowledge in-thread.
- Check content, identity, author, destination, uniqueness and all unchanged workspace state.
- Wrong thread, replacement instead of edit, stale fact or collateral change fails.



- Exact text is deliberate. Fully undone actions may pass; path-level safety is not currently graded.

Verification separates software checks from model runs

- 169 backend/harness checks and 90 browser checks.
- 84 valid states and 2,583 adversarial invalid states for the 12 deep-task graders.
- Scripted UI recipes exercise interaction paths; they do not measure model capability.
- 249 / 306 recorded model/task attempts in the one-pass comparison; earlier interface trials remain separate.



Software tests
Does the app work?



Grader challenges
Does reward mean success?



Model episodes
Can the policy do the task?

One attempt per model and task

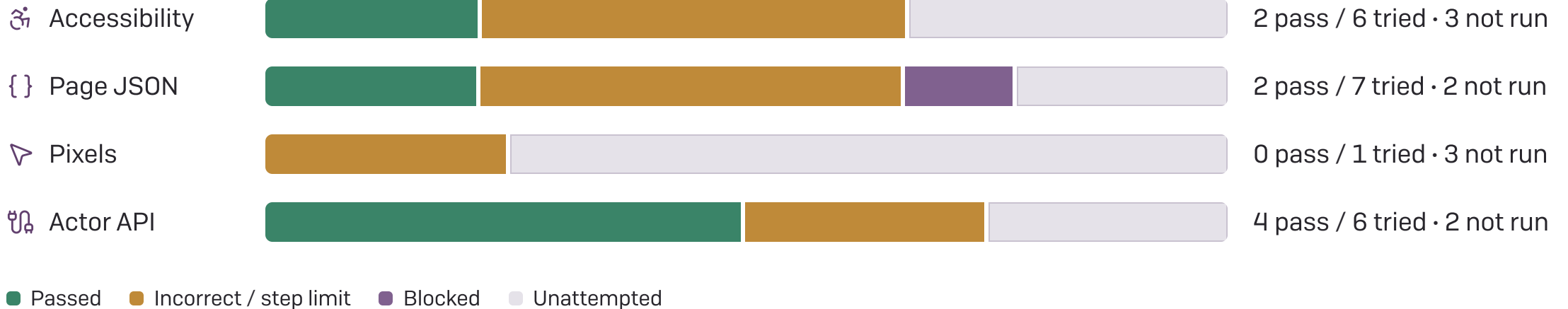
 17 model routes 18 tasks 1 attempt per cell

- Flagship, standard and efficient routes across nine model families.
- Fresh workspaces; seed 1042; accessibility control; recent-four history; no guide.
- 40 actions; 4,096 output tokens per call; 180 seconds and \$5 maximum per trial.
- 105 prior attempts retained; only 201 untouched cells continued. No repeats or dropped failures.
- \$300 cumulative ceiling, including prior pilots. A trial spend stop stays blocked.
- Public tasks and temporal cohorts: coverage evidence, not a reliable model ranking.

-
- [Frozen plan, preserved origins and explicit scheduling amendment.](#)

Earlier interface study · separate from the 306 cells

- 20 attempted: 8 passed, 11 incorrect/step-limited, 1 blocked; 10 unattempted.
- 126 requests; \$0.149874 estimated including \$0.001490 reserved for the blocked response.



- Unequal task/model coverage; these totals are not causal interface comparisons. Cost is not an invoice.
- [Complete inventory, exact trajectories and hashes.](#)

17 models. Every task.

1 per task/model · Collection stopped

249 / 306 attempted · 123 passed · 50 incomplete · 76 blocked

Model ↕	Trials ↕	Passed ↕	Incomplete ↕	Blocked ↕	Median time ↕	Est. cost ↕
 GPT-6 Astra	14 / 18	11 / 14	1	2	27.6s	\$27.156
 GPT-6.1 Sol	15 / 18	12 / 15	3	0	27.2s	\$6.904
 GPT-6 Luna	15 / 18	7 / 15	7	1	60.2s	Unknown \$0.507 incl. reservation
 GPT-OSS 120B	14 / 18	4 / 14	2	8	180.1s	Unknown \$0.613 incl. reservation
 Claude Fable 5.1	15 / 18	11 / 15	1	3	64.0s	\$39.708
 Claude Opus 5.5	14 / 18	10 / 14	3	1	38.0s	Unknown \$22.980 incl. reservation
 Claude Sonnet 5.5	15 / 18	6 / 15	4	5	53.6s	Unknown \$13.870 incl. reservation
 Claude Haiku 4.5	14 / 18	5 / 14	2	7	28.3s	\$3.048
 Grok 4.7	14 / 18	7 / 14	5	2	32.8s	Unknown \$9.985 incl. reservation
 Qwen 3.8 Max	15 / 18	6 / 15	1	8	121.6s	Unknown \$6.648 incl. reservation
 DeepSeek V4 Pro	15 / 18	6 / 15	0	9	108.6s	Unknown \$4.349 incl. reservation
 DeepSeek V4.1 Flash	15 / 18	6 / 15	3	6	57.2s	Unknown \$1.161 incl. reservation
 GLM 5.3	14 / 18	6 / 14	0	8	118.9s	Unknown \$3.028 incl. reservation
 GLM 5.3 Flash	16 / 18	7 / 16	0	9	46.2s	\$0.395
 Kimi K3	15 / 18	8 / 15	3	4	66.7s	Unknown \$13.649 incl. reservation
 MiniMax M3	14 / 18	5 / 14	6	3	24.7s	Unknown \$0.955 incl. reservation
 Nemotron 3 Ultra	15 / 18	6 / 15	9	0	55.0s	\$3.662

Passed Incomplete Blocked Not run

- 18 public tasks · 1 attempt per task/model · accessibility control · \$300 shared ceiling. Snapshot 2026-10-03 19:46 UTC.
- 105 prior attempts preserved, including failures. New cells only; no retries. Family/tier coverage, not a popularity ranking. Raw outcomes; release-sync and design-handoff have wording caveats.
- [Frozen design](#) · [Collection status and evidence](#) · [Download all trials \(CSV\)](#) · [Earlier 36-trial pilot: 20 passed](#)

What the current trajectories explain

- Thread repair: 13 / 17 passed; handoff DM: 16 / 17. Thread repair combines fact retrieval, an original-message edit, a reaction and a reply.
- Saved-item cleanup: 4 / 17 strict passes. A further diagnostic workspace pass ended at an output limit and remains blocked.
- Design handoff: 0 / 17. Six completed runs copied the quoted “DESIGN”; the grader expected the retrieved name. The substitution instruction was implicit.
- Release-sync: 0 / 17, with an ambiguous source-location hint—not clean capability evidence. Sol made partial changes, then repeated edit/cancel actions.

-
- [Trace identities and outcome details](#). One attempt per public task; no claim of repeatability or a causal model ranking.

Speed and resource measurements



17.34 ms

Create session · p50



1.50 ms

Reset state · p50

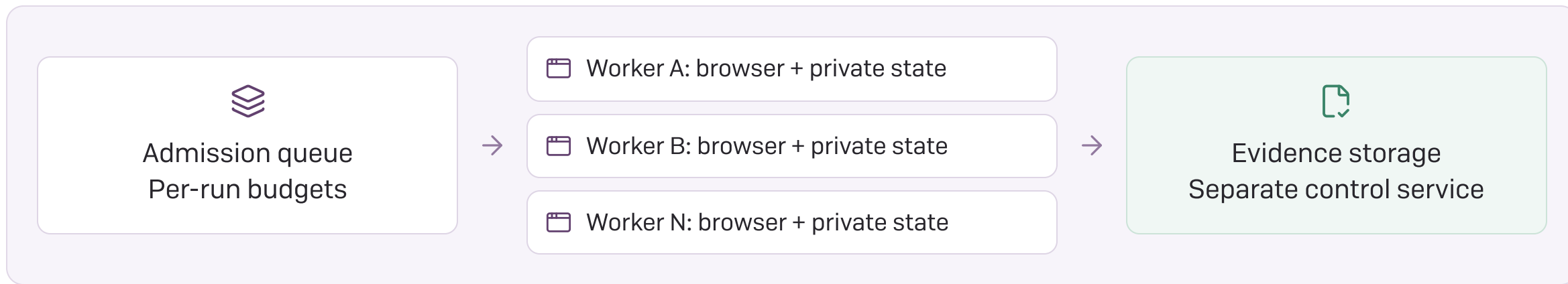


84.15 ms

Screenshot · p50

- 100 persisted sessions, 400 reads at concurrency 16, and 10 sequential browser contexts.
 - Machine: Apple M5 Max, Node 24; timings include the stated operation only.
 - Browser child-process memory, sustained agent concurrency and multi-host capacity are not measured.
 - Model calls and visual evidence add latency; API throughput is not agent throughput.
-
- [Raw samples, machine details and measurement boundaries.](#)

Scaling path and tradeoffs



- Proposed architecture, not a measured deployment at scale.
- Partition synchronous SQLite work; pool bounded browser processes and recycle workers.
- Use containers/VMs and network isolation before allowing shell-capable or hostile policies.
- Measure full worker memory, p95 latency, failure rate and cleanup under a sustained load.

Demonstration and discussion

- [Open the workspace](#): search, reply in a thread, and edit your own message.
 - [Open trial review](#): choose a recorded model/task, play a success, and inspect an incomplete or blocked attempt.
 - Inspect the exact request, action, state change and terminal checks; replay requires no model calls.
 - Discuss: required Slack fidelity, observation contract, security boundary and cost/throughput target.
 - Next gates: successful pixel policies, task-template holdouts, trainer integration and concurrent-worker measurements.
-
- Reset/step/reward are implemented; no RL training or BrowserGym/OpenEnv compatibility is claimed.
 - [Code and setup](#) · [Submission assessment and limitations](#).